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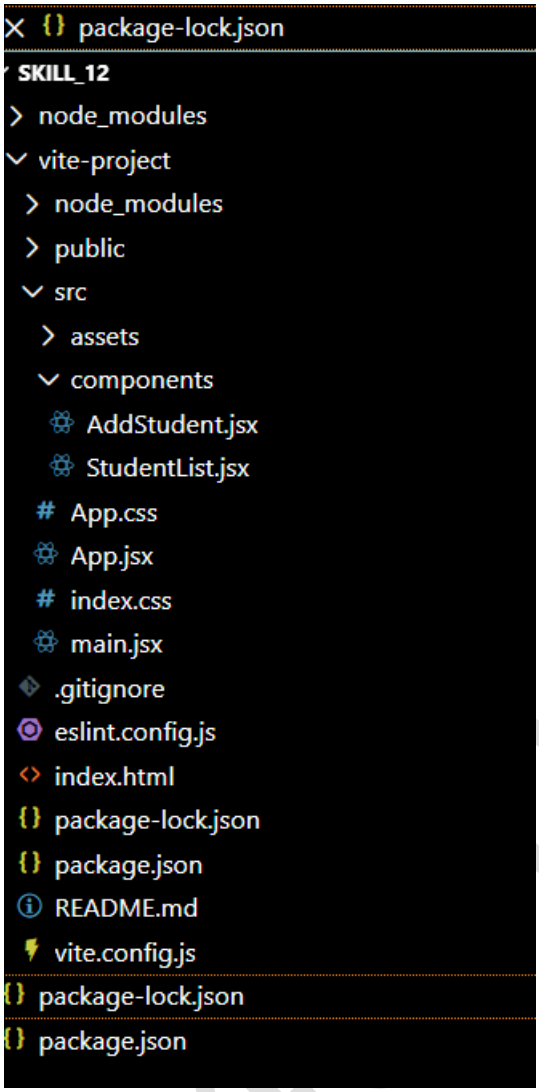
ID:2400030211
Section:59

Project structure:



StudentList.jsx

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```jsx
import React, { useEffect, useState } from 'react';
import axios from 'axios';

const StudentList = ({ onEdit }) => {
    const [students, setStudents] = useState([]);

    // Function to fetch students
    const fetchStudents = async () => {
        try {
            const response = await axios.get("http://localhost:8080/students");
            setStudents(response.data);
        } catch (error) {
            console.error("Error fetching students:", error);
        }
    };

    useEffect(() => {
        fetchStudents();
    }, []);

    const deleteStudent = async (id) => {
        await axios.delete(`http://localhost:8080/students/${id}`);
```

AddStudent.jsx

```jsx
import React, { useState, useEffect } from 'react';
import axios from 'axios';

const AddStudent = ({ selectedStudent, refreshList, clearSelection }) => {
    const [student, setStudent] = useState({ name: '', email: '', course: '' });

    // Fill form if editing
    useEffec (parameter) selectedStudent: any
        if (selectedStudent) setStudent(selectedStudent);
    }, [selectedStudent]);

    const handleChange = (e) => {
        setStudent({ ...student, [e.target.name]: e.target.value });
    };

    const handleSubmit = async (e) => {
        e.preventDefault();
        if (student.id) {
            // Update
            await axios.put(`http://localhost:8080/students/${student.id}`, student);
        } else {
            // Create
            await axios.post("http://localhost:8080/students", student);
```

App.jsx

```jsx
import React, { useState } from 'react';
import StudentList from './components/StudentList';
import AddStudent from './components/AddStudent';

function App() {
    // Stores the student currently being edited
    const [selectedStudent, setSelectedStudent] = useState(null);

    // A simple toggle to force the StudentList to refresh data
    const [refreshKey, setRefreshKey] = useState(0);

    const handleRefresh = () => {
        setRefreshKey(prev => prev + 1);
    };

    const handleEdit = (student) => {
        setSelectedStudent(student);
        // Scroll to top so user sees the form is filled
        window.scrollTo(0, 0);
    };

    return (
        <div style={{
```

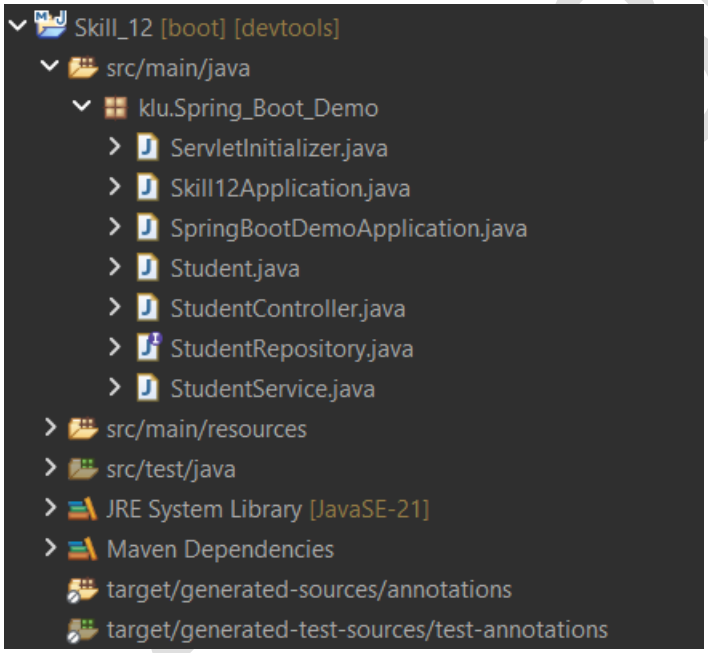
Package.json

```json
{
  "name": "vite-project",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  ▷Debug
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
    "preview": "vite preview"
  },
  "dependencies": {
    "axios": "^1.6.7",
    "react": "^18.2.0",
    "react-dom": "^18.2.0"
  },
  "devDependencies": {
    "@eslint/js": "^9.9.0",
    "@types/react": "^18.3.3",
    "@types/react-dom": "^18.3.0",
    "@vitejs/plugin-react": "^4.3.1",
    "eslint": "^9.9.0",
```

Packagelock.json:


```
{
  "name": "skill_12",
  "lockfileVersion": 3,
  "requires": true,
  "packages": {
    "": {
      "dependencies": {
        "axios": "^1.13.6"
      }
    },
    "node_modules/asynckit": {
      "version": "0.4.0",
      "resolved": "https://registry.npmjs.org/asynckit/-/asynckit-0.4.0.tgz",
      "integrity": "sha512-Oei9OH4tRh0YqU3GxhX79dM/mwVgvbZJaSNaRk+bshkj0S5cfHcgYakreBjrHwatXKb
      "license": "MIT"
    },
    "node_modules/axios": {
      "version": "1.13.6",
      "resolved": "https://registry.npmjs.org/axios/-/axios-1.13.6.tgz",
      "integrity": "sha512-ChTCHMouEe2kn713WHbQGcuYrr6fXTBiu460OTwWrWob16g1bXn4vtz07Ope7ewMozJ
      "license": "MIT",
      "dependencies": {
        "follow-redirects": "^1.15.11",
```


Project structure:



Student.java

```java
package klu.Spring_Boot_Demo;

import jakarta.persistence.*;

@Entity
@Table(name = "student_table")
public class Student {
    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;
    private String name;
    private String email;
    private String course;

    // Getters and Setters
    public Long getId() { return id; }
    public void setId(Long id) { this.id = id; }
    public String getName() { return name; }
    public void setName(String name) { this.name = name; }
    public String getEmail() { return email; }
    public void setEmail(String email) { this.email = email; }
    public String getCourse() { return course; }
    public void setCourse(String course) { this.course = course; }
}
```

StudentController.java

```java
package klu.Spring_Boot_Demo;

import org.springframework.beans.factory.annotation.Autowired;

@RestController
@RequestMapping("/students")
@CrossOrigin(origins = "http://localhost:5173") // Connects to your React app
public class StudentController {

    @Autowired
    private StudentService service;

    @GetMapping
    public List<Student> getAll() {
        return service.getAllStudents();
    }

    @PostMapping
    public ResponseEntity<Student> add(@RequestBody Student student) {
        return ResponseEntity.ok(service.saveStudent(student));
    }

    @PutMapping("/{id}")
    public ResponseEntity<Student> update(@PathVariable Long id, @RequestBody Student studentDetails) {
        Student student = service.getStudentById(id);
        if (student != null) {
            student.setName(studentDetails.getName());
            student.setEmail(studentDetails.getEmail());
            student.setCourse(studentDetails.getCourse());
            return ResponseEntity.ok(service.saveStudent(student));
        }
```

StudentRepository.java

```java
package klu.Spring_Boot_Demo;

import org.springframework.data.jpa.repository.JpaRepository;

@Repository
public interface StudentRepository extends JpaRepository<Student, Long> {
}
```

StudentService.java

```java
package klu.Spring_Boot_Demo;

import org.springframework.beans.factory.annotation.Autowired;

@Service
public class StudentService {
    @Autowired
    private StudentRepository repository;

    public List<Student> getAllStudents() {
        return repository.findAll();
    }

    public Student saveStudent(Student student) {
        return repository.save(student);
    }

    public void deleteStudent(Long id) {
        repository.deleteById(id);
    }

    public Student getStudentById(Long id) {
        return repository.findById(id).orElse(null);
    }
}
```

SpringBootDemoApplication.java

```java
package klu.Spring_Boot_Demo;

import org.springframework.boot.SpringApplication;

@SpringBootApplication
public class SpringBootDemoApplication {
    public static void main(String[] args) {
        // This line tells Spring Boot to start the whole application
        SpringApplication.run(SpringBootDemoApplication.class, args);
    }
}
```

Skill12Application.java

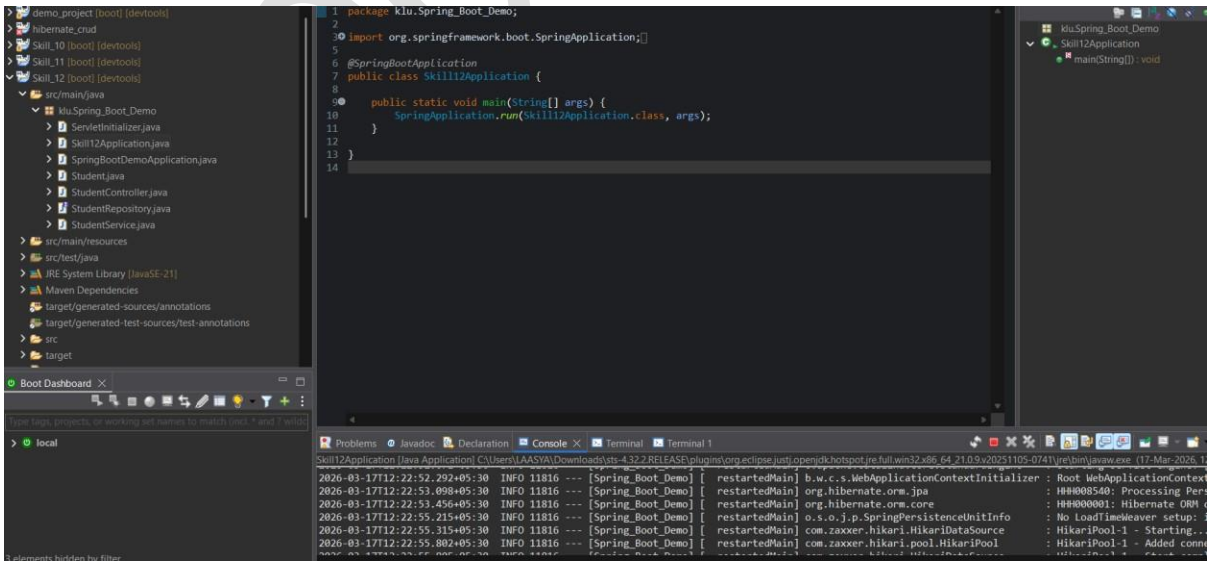
```java
package klu.Spring_Boot_Demo;

import org.springframework.boot.SpringApplication;

@SpringBootApplication
public class Skill12Application {

    public static void main(String[] args) {
        SpringApplication.run(Skill12Application.class, args);
    }

}
```



```json
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  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint .",
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    "react-dom": "^18.2.0"
  },
  "devDependencies": {
    "@eslint/js": "^9.9.0",
    "@types/react": "^18.3.3",
    "@types/react-dom": "^18.3.0",
    "@vitejs/plugin-react": "^4.3.1",
    "eslint": "^9.9.0",
```

OUTPUT:

24000302

Submit

ktudent List

Name	Email	Course	Actions

GitHub Link: https://github.com/hemalatharachakonda/2400030211-Skill_experiments.git

2400030211